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INFLUENCING HUMANS TO CONFORM TO PREFERENCE MODELS FOR RLHF

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ABSTRACT

Designing a reinforcement learning from human feedback (RLHF) algorithm to approximate a human's unobservable reward function requires assuming, implicitly or explicitly, a model of human preferences. A preference model that poorly describes how humans generate preferences risks learning a poor approximation of the human's reward function. In this paper, we conduct three human studies to assess whether one can influence the expression of real human preferences to more closely conform to a desired preference model. Importantly, our approach does not seek to alter the human's unobserved reward function. Rather, we change how humans use this reward function to generate preferences, such that they better match whatever preference model is assumed by a particular RLHF algorithm. We introduce three interventions: showing humans the quantities that underlie a preference model, which is normally unobservable information derived from the reward function; training people to follow a specific preference model; and modifying the preference elicitation question. All intervention types show significant effects, providing practical tools to improve preference data quality and the resultant alignment of learned reward functions. Overall we establish a novel research direction in model alignment: designing interfaces and training interventions to increase human conformance with the modeling assumptions of the algorithm that will learn from their input.

1 Introduction

Aligning agent behavior with human preferences is a central goal of reinforcement learning from human feedback (RLHF). This process generally assumes a model of human preferences, a probability distribution over a human's rankings of pairs of trajectory segments based on their reward function. The RLHF algorithm cannot observe the human's reward function and instead must approximate it from preferences.

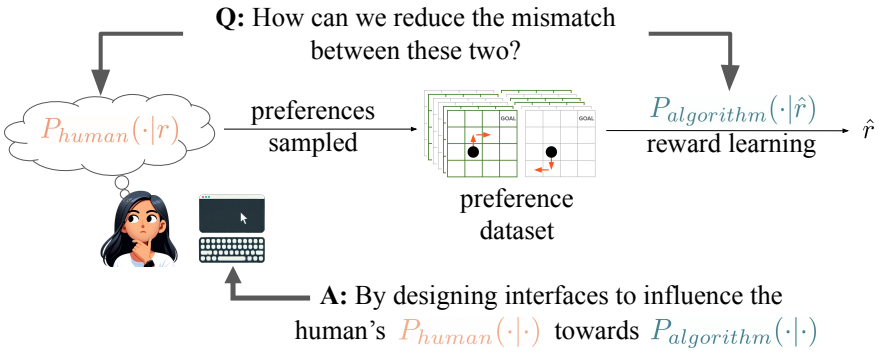

Figure 1: Our proposed method of influencing human preferences. We design interfaces to influence the human's preferences without changing their underlying reward function.

Prior work has explored different choices for this preference model and provided evidence that the more aligned the RLHF algorithm's preference model is with how humans generate preferences, the more aligned the learned reward function is (Knox et al., 2022). Most work assumes that human preferences arise probabilistically from *partial return*, the sum of rewards over a trajectory segment. By this assumption, humans presented with two trajectory segments tend to prefer the one that accrues greater reward, as measured by their reward function (Christiano et al.,

2017; Ouyang et al., 2022). Other work has challenged whether partial return is a sufficiently descriptive model of human preferences and proposed alternative models of human preference (Knox et al., 2022; Kim et al., 2023). We instead take a *prescriptive* approach to learning from human preferences. Specifically, we propose designing training and preference elicitation interfaces to influence humans to better conform to a chosen preference model. We test influencing humans towards the partial return preference model or *regret* preference model, which is based on each segment's deviation from optimality (Knox et al., 2022). See Figure 1 for a visual summary of our interventions.

We introduce three methods and evaluate how each method can influence humans towards these two different preference models. First, in the PRIVILEGED experiment we present subjects with calculations of regret or partial return during preference elicitation, thus providing the information needed to exactly follow the *target preference model*. This first intervention is a proof of concept, since it requires knowledge of the reward function that is untenable in organic preference elicitation. In contrast, the next two methods can be deployed in practice. In the TRAINED experiment we train humans to follow a specific preference model. In the QUESTION experiment we change *only* the QUESTION asked during preference elicitation. The PRIVILEGED and TRAINED methods result in close conformance with either target preference model. The QUESTION method also increases conformance with either preference model, but significant effects are only observed when targeting the partial return preference model.

At a high level, we contribute a straightforward approach to improving model alignment: training humans and designing interfaces to increase human conformance with the assumptions of the algorithm that learns from their input. Our experiments yield guidance for RLHF practitioners and lay groundwork for future research at the intersection of interface design and learning from human input, RLHF especially. The code for all computational experiments and interfaces for collecting preferences, as well as the collected datasets of human preferences, are available here.

2 Related Work: Learning from and Modeling Human Preferences

Extensive research has explored learning from human preferences for RLHF. This research includes RLHF approaches that explicitly learn a reward function (Christiano et al., 2017; Ibarz et al., 2018; Sadigh et al., 2017; Lee et al., 2021a,b; Ziegler et al., 2019; Ouyang et al., 2022; OpenAI, 2022; Bıyık et al., 2022; Wang et al., 2022; Bai et al., 2022; Glaese et al., 2022; Knox et al., 2022; Touvron et al., 2023; Ethayarajh et al., 2024) and other approaches that directly learn a policy or advantage function from human preferences (Rafailov et al., 2024; Knox et al., 2024; Hejna et al., 2023). All of the algorithms in the works cited above assume one of the two models of human preference that this paper uses to design preference elicitation interventions. Our investigation of how to increase human conformance with the assumed preference model is compatible with and strengthens this past research.

Other research has sought agent alignment by developing preference models that better model human preferences, such as by assuming that human preferences arise from a segment's regret (Knox et al., 2022) or weighted sum of non-Markovian rewards (Kim et al., 2023), instead of a segment's sum of Markovian rewards as is commonly assumed. However, improving the preference model still results in some gap between the preference model and actual human preferences, which are subject to difficult-to-model confounding factors and individual differences. Our research seeks to close this gap for whatever preference model is chosen.

Further, even if we had a *perfect* descriptive model of how *all* humans generate preferences, we may want preferences to be generated by a different preference model that permits more tractable algorithms, greater sample efficiency, or some theoretical guarantees. The methods we introduce can also help in these settings.

3 Preliminaries: Preference Model Choices

When influencing human preferences, we analyze two preference models: partial return and regret. In this section, we explain the assumptions encoded in each preference model and how each can be used in RLHF.

Consider a Markov decision process (MDP) that represents the task environment using a tuple $(S, A, T, \gamma, D_0, r)$. S and A are the sets of possible states and actions, respectively. $T : S \times A \to p(\cdot|s, a)$ is a transition function; γ is the discount factor; and D_0 is the distribution of start states. Unless stated otherwise, we assume tasks are undiscounted ($\gamma = 1$) and have terminal states, after which only 0 reward can be received. r is a reward function, $r : S \times A \times S \to \mathbb{R}$, where r_t is a function of s_t, a_t, and s_{t+1} at time t. An MDP$\backslash r$ is an MDP without a reward function.

Let r refer to the ground-truth reward function for some MDP, $\hat{r}$ refer to a learned approximation of r, and $\tilde{r}$ refer to any reward function (including r or $\hat{r}$). A policy ($\pi : S \times A \to [0, 1]$) specifies the probability of an action given a state. $Q^{\pi}_{\tilde{r}}$ and $V^{\pi}_{\tilde{r}}$ refer respectively to the state-action value function and state value function for a policy, π, under $\tilde{r}$, and are defined as follows: $V^{\pi}_{\tilde{r}}(s) \triangleq \mathbb{E}_{\pi}[\sum_{t=0}^{\infty} \tilde{r}(s_t, a_t, s_{t+1})|s_0 = s]$ and $Q^{\pi}_{\tilde{r}}(s, a) \triangleq \mathbb{E}_{\pi}[\tilde{r}(s, a, s') + V^{\pi}_{\tilde{r}}(s')]$.

An optimal policy $\pi^*_{\tilde{r}}$ is any policy where $V^{\pi^*_{\tilde{r}}}_{\tilde{r}}(s) \geq V^{\pi}_{\tilde{r}}(s)$ at every state s for every policy π. We write shorthand for $Q^{\pi^*_{\tilde{r}}}_{\tilde{r}}$ and $V^{\pi^*_{\tilde{r}}}_{\tilde{r}}$ as $Q^*_{\tilde{r}}$ and $V^*_{\tilde{r}}$, respectively.

3.1 Reward Learning from Pairwise Preferences

RLHF typically learns a reward function by minimizing the cross-entropy loss—i.e., maximizing the likelihood—of observed human preference labels (Christiano et al., 2017; Ibarz et al., 2018; Wang et al., 2022; Bıyık et al., 2021; Sadigh et al., 2017; Lee et al., 2021a,b; Ziegler et al., 2019; Ouyang et al., 2022; Bai et al., 2022; Glaese et al., 2022; OpenAI, 2022; Touvron et al., 2023; Hejna III & Sadigh, 2023). Here, we examine the preliminaries needed to precisely define and this loss function when learning from preferences.

Segments Let σ denote a segment starting at state s^σ_0. Its length $|\sigma|$ is the number of transitions within the segment. A segment includes $|\sigma|+1$ states and $|\sigma|$ actions: $(s^\sigma_0, a^\sigma_0, s^\sigma_1, a^\sigma_1, ..., s^\sigma_{|\sigma|})$. In this problem setting, segments lack any reward information. As shorthand, we define $\sigma_t \triangleq (s^\sigma_t, a^\sigma_t, s^\sigma_{t+1})$. Additionally, we denote the *partial return* of a segment σ as $\Sigma_\sigma \tilde{r}$ for some $\tilde{r}$, where $\tilde{r}^\sigma_t \triangleq \tilde{r}(s^\sigma_t, a^\sigma_t, s^\sigma_{t+1})$ and $\Sigma_\sigma \tilde{r} \triangleq \sum_{t=0}^{|\sigma|-1} \tilde{r}^\sigma_t$.

Preference datasets We denote a preference dataset as $D_\succ$, which comprises samples of preferences over pairs of segments. Each sample is represented as $(\sigma_1, \sigma_2, \mu)$. Vector $\mu = \langle \mu_1, \mu_2 \rangle$ represents the preference; specifically, if σ_1 is preferred over σ_2, denoted $\sigma_1 \succ \sigma_2$, $\mu = \langle 1, 0 \rangle$. μ is $\langle 0, 1 \rangle$ if $\sigma_1 \prec \sigma_2$ and is $\langle 0.5, 0.5 \rangle$ for $\sigma_1 \sim \sigma_2$ (no preference). For a sample $(\sigma_1, \sigma_2, \mu)$, we assume that the two segments have equal lengths (i.e., $|\sigma_1| = |\sigma_2|$) and the same start state (i.e., $s^{\sigma_1}_0 = s^{\sigma_2}_0$).

Loss function When learning a reward function from a preference dataset, $D_\succ$, preference labels are typically assumed to be generated by a preference model P based on an unobservable *ground-truth* reward function r. We learn $\hat{r}$, an approximation of r, by minimizing this cross-entropy loss:

$$loss(\hat{r}, D_\succ) = -\sum_{(\sigma_1,\sigma_2,\mu)\in D_\succ} \mu_1 \log P(\sigma_1 \succ \sigma_2|\hat{r}) + \mu_2 \log P(\sigma_1 \prec \sigma_2|\hat{r}) \tag{1}$$

If $\sigma_1 \succ \sigma_2$, the sample's likelihood is $P(\sigma_1 \succ \sigma_2|\hat{r})$ and its loss is therefore $-logP(\sigma_1 \succ \sigma_2|\hat{r})$. If $\sigma_1 \prec \sigma_2$, its likelihood is $1 - P(\sigma_1 \succ \sigma_2|\hat{r})$. This loss is under-specified until the preference model $P(\sigma_1 \succ \sigma_2|\hat{r})$ is defined. Learning approximations of r from preferences can be summarized as "minimize Equation 1".

Preference models A preference model determines the likelihood of one trajectory segment being preferred over another, $P(\sigma_1 \succ \sigma_2|\tilde{r})$.

3.2 Preference Models: Partial Return and Regret

Here we describe the two preference models that are most commonly used when learning from human preferences. Figure 2 illustrates an example of how these two models differ. In Section 5 we detail our proposed training procedures and preference elicitation interfaces that influence human preferences to conform to either choice of preference model. We focus on two preference models that differ solely in the segment statistic they use to evaluate the desirability of a trajectory segment. We do not address other aspects of modeling human preferences, such as different assumptions about human irrationality, risk aversion, or uncertainty. However, the interventions proposed in this paper, which aim to influence humans towards a chosen preference model, may be generalizable to influencing humans to conform to preference models that incorporate these additional factors.

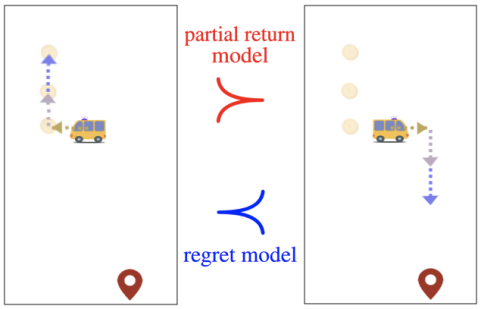

Figure 2: On each step, the vehicle receives reward as the sum of reward components: -1 for every time it moves; $+1$ for collecting a coin; and $+50$ for reaching the red goal marker, which ends the episode. The partial return preference model favors the left trajectory, while regret favors the right.

Partial return The most common preference model (e.g., Christiano et al. (2017)) posits that human preferences are generated by a Boltzmann distribution over the partial returns of the two segments, expressed here as a logistic function:

$$P_{\Sigma_r}(\sigma_1 \succ \sigma_2|\tilde{r}) = logistic\Big(\Sigma_{\sigma_1}\tilde{r} - \Sigma_{\sigma_2}\tilde{r}\Big). \tag{2}$$

Regret An alternative model of human preferences is based on regret (Knox et al., 2022). This model suggests that human preferences arise from the deviations of each segment from optimal decision-making, characterized by the regret